

```

[ > restart;
[ > f1 := x^2 - y^2 - 1;
[                                      $f1 := x^2 - y^2 - 1$ 
[ > f2 := x^3 - 3*x*y^2 + 3*x^2*y - y^3 + 1;
[                                      $f2 := x^3 - 3xy^2 + 3x^2y - y^3 + 1$ 
[ > alias( alpha=RootOf( 6*y+3+6*y^3+2*y^2, y ) );
[                                      $\alpha$ 
[ > solve( {f1, f2}, {x,y} );
[                                      $\{y=0, x=-1\}, \{y=\alpha, x=-\frac{3}{7}\alpha + \frac{6}{7}\alpha^2 + \frac{5}{7}\}$ 
[ > with(Groebner);
[ [MulMatrix, SetBasis, ToricIdealBasis, fglm_algo, gbasis, gsolve, hilbertdim, hilbertpoly,
[   hilbertseries, inter_reduce, is_finite, is_solvable, leadcoeff, leadmon, leadterm, normalf,
[   pretend_gbasis, reduce, spoly, termorder, testorder, univpoly]
[ > gb1 := gbasis( {f1,f2}, plex(x,y) );
[                                      $gb1 := [6y^2 + 6y^4 + 2y^3 + 3y, 24y^3 + 27y + 7 + 7x + 2y^2]$ 
[ > factor(gb1[1]);
[                                      $y(6y + 3 + 6y^3 + 2y^2)$ 
[ > g1 := x^3*y^2 + c1*x^3*y + y^2 + c2*x + c3;
[                                      $g1 := x^3y^2 + c1x^3y + y^2 + c2x + c3$ 
[ > g2 := c4*x^4*y^2 - x^2*y + y + c5;
[                                      $g2 := c4x^4y^2 - x^2y + y + c5$ 
[ > gb2 := gbasis({g1,g2}, tdeg(x,y)):
[
[ Change the : in the last problem for a ; to see the output
[ > h1 := c*x[1]^2*x[2] + 9*x[1]^2+2*x[1]*x[2] + 5*x[1] + x[2] - 3;
[                                      $h1 := cx_1^2x_2 + 9x_1^2 + 2x_1x_2 + 5x_1 + x_2 - 3$ 
[ > h2 := 2*x[1]^3*x[2] + 6*x[1]^3 - 2*x[1]^2 - x[1]*x[2] - 3*x[1] -
[   x[2] + 3;
[                                      $h2 := 2x_1^3x_2 + 6x_1^3 - 2x_1^2 - x_1x_2 - 3x_1 - x_2 + 3$ 
[ > h3 := x[1]^3*x[2] + 3*x[1]^3 + x[1]^2*x[2] + 2*x[1]^2;
[                                      $h3 := x_1^3x_2 + 3x_1^3 + x_1^2x_2 + 2x_1^2$ 
[ > infolevel[primpart] := 3;
[                                      $infolevel_{primpart} := 3$ 
[ > gb3 := gbasis( [h1, h2, h3 ], tdeg(x[1],x[2]) );
[ Groebner/primpartscale: "remove content" 1
[ Groebner/primpartscale: "total degree drops from" 4 " to " 4
[ Groebner/primpartscale: "remove content" 1
[ Groebner/primpartscale: "total degree drops from" 4 " to " 4
[ Groebner/primpartscale: "remove content" 1
[ Groebner/primpartscale: "total degree drops from" 4 " to " 4

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Groebner/primpartscale: "remove content" 1
Groebner/primpartscale: "total degree drops from" 3 " to " 3
Groebner/primpartscale: "remove content" 1
Groebner/primpartscale: "total degree drops from" 5 " to " 5
Groebner/primpartscale: "remove content" 1
Groebner/primpartscale: "total degree drops from" 2 " to " 2
Groebner/primpartscale: "remove content" 1
Groebner/primpartscale: "total degree drops from" 3 " to " 3
Groebner/primpartscale: "remove content" 769034410057
Groebner/primpartscale: "total degree drops from" 3 " to " 3
Groebner/primpartscale: "remove content" c
Groebner/primpartscale: "total degree drops from" 5 " to " 4
Groebner/primpartscale: "remove content" 13025984346677562934802
Groebner/primpartscale: "total degree drops from" 2 " to " 2
Groebner/primpartscale: "remove content" c^2
Groebner/primpartscale: "total degree drops from" 5 " to " 3
Groebner/primpartscale: "remove content" 3630762687546
Groebner/primpartscale: "total degree drops from" 2 " to " 2
Groebner/primpartscale: "remove content" 6*c-18
Groebner/primpartscale: "total degree drops from" 5 " to " 4
Groebner/primpartscale: "remove content" 9171655931547777555168750
Groebner/primpartscale: "total degree drops from" 1 " to " 1
Groebner/primpartscale: "remove content" 54-36*c+6*c^2
Groebner/primpartscale: "total degree drops from" 5 " to " 3
Groebner/primpartscale: "remove content" 4875192059527403597039811141789551739
86259810819684800
Groebner/primpartscale: "total degree drops from" 1 " to " 1
Groebner/primpartscale: "remove content" 47316150*c^2+75557300+771750*c^4-9825
3750*c-9969050*c^3
Groebner/primpartscale: "total degree drops from" 5 " to " 1
Groebner/primpartscale: "remove content" 252*c^2-1643*c+2966
Groebner/primpartscale: "total degree drops from" 3 " to " 1
Groebner/primpartscale: "remove content" 1
Groebner/primpartscale: "total degree drops from" 1 " to " 1
Groebner/primpartscale: "remove content" 1
Groebner/primpartscale: "total degree drops from" 1 " to " 1
Groebner/primpartscale: "remove content" 1
Groebner/primpartscale: "total degree drops from" 1 " to " 1

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$$gb3 := [x_2 - 3, x_1]$$

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> infolevel[primpart] := 0;
                                     infolevelprimpart := 0
> gb4 := gbasis( [ eval(h1,c=3), h2, h3 ], tdeg(x[1],x[2]) );
                                     gb4 := [2 x22 - 5 x2 - 8 x1 - 3, x1 x2 + x1 - x2 + 3, 2 x12 - 3 x1 + 2 x2 - 6]
>

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